LUMS
A Not-for-Profit University

School of Science and Engineering

AI 501 Mathematics for Artificial Intelligence

ASSIGNMENT 2 – SOLUTIONS

Due Date: 1 pm, Friday, October 31, 2025.
Format: 5 problems for a total of 100 marks
Instructions:

- You are allowed to collaborate with your peers but copying your colleague's solution is strictly prohibited. This is not a group assignment. Each student must submit his/her own assignment.
- Solve the assignment on blank A4 sheets and staple them before submitting.
- Submit in-class or in the dropbox labeled AI-501 outside the instructor's office.
- **Write your name and roll no. on the first page.**
- Feel free to contact the instructor or the teaching assistants if you have any concerns.

- You represent the most competent individuals in the country, do not let plagiarism come in between your learning. In case any instance of plagiarism is detected, the disciplinary case will be dealt with according to the university's rules and regulations.
- We require you to acknowledge any use or contributions from generative AI tools. Include the following statement to acknowledge the use of AI where applicable.

  I have used [insert Tool Name] to [write, generate, plot or compute; explain specific use of generative AI] [number of times].

Problem 1 (20 marks)

Polynomial Regression and the Bias-Variance Trade-off

Consider a dataset with a single feature x and a target variable y. We model the relationship using polynomial regression of degree M.

(a) For a given data point x_i, the prediction is

$$\hat{y}_i = \theta_0 + \theta_1 x_i + \theta_2 x_i^2 + \cdots + \theta_M x_i^M.$$

Given n data points, write the vectorized form of the model $y = A\theta$. Define the design matrix A and explain how its structure and condition number are influenced by the polynomial degree M.

(b) The regularized cost function for Ridge regression is given by:

$$J(\theta) = \frac{1}{2n}\|y - A\theta\|_2^2 + \frac{\lambda}{2}\|\theta\|_2^2.$$

Derive the closed-form solution for the parameter vector θ that minimizes $J(\theta)$.

(c) Explain the impact of a very large value of the regularization parameter λ on the bias and variance of the polynomial regression model.

(d) How does the choice of M interact with λ in this context?

Solution:

(a) The vectorized form is $\mathbf{y} = A\theta$, where:

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^M \\ 1 & x_2 & x_2^2 & \dots & x_2^M \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^M \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_M \end{bmatrix}$$

As M increases:

- The columns of A become higher powers of x_i
- Columns become nearly linearly dependent for large M (since x^m and x^{m+1} are highly correlated)
- This makes A^TA nearly singular $\rightarrow$ condition number $\kappa(A^TA)$ becomes very large
- High condition number means numerical instability and sensitivity to small changes in data

(b) Starting from the cost function:

$$J(\theta) = \frac{1}{2n}\|y - A\theta\|_2^2 + \frac{\lambda}{2}\|\theta\|_2^2$$

Multiply by $2n$:

$$2nJ(\theta) = (y - A\theta)^T(y - A\theta) + n\lambda\theta^T\theta$$

Expand and take derivative w.r.t. θ:

$$\frac{\partial}{\partial\theta}\left[y^Ty - 2y^TA\theta + \theta^TA^TA\theta + n\lambda\theta^T\theta\right] = 0$$

$$-2A^Ty + 2A^TA\theta + 2n\lambda\theta = 0$$

$$A^TA\theta + n\lambda\theta = A^Ty$$

$$(A^TA + n\lambda I)\theta = A^Ty$$

Thus, the closed-form solution is:

$$\theta = (A^TA + n\lambda I)^{-1}A^Ty$$

(c) Impact of large λ:

- **Bias:** Increases - model becomes too constrained to fit data well
- **Variance:** Decreases - parameters become less sensitive to training data variations
- As $\lambda \rightarrow \infty$, $\theta \rightarrow 0 \rightarrow$ model approaches constant prediction

(d) Interaction between M and λ:

- High M without regularization $\rightarrow$ high variance (overfitting), low bias
- High M with very large $\lambda \rightarrow$ high bias (underfitting), low variance
- There's a trade-off: for given M, increasing λ reduces variance but increases bias
- If M is large, need more regularization (λ large) to control variance
- If M is small, even small λ might suffice; large λ causes underfitting quickly

Optimal performance occurs at intermediate λ for given M, balancing model flexibility and constraint.

Problem 2 (20 marks)

Comparing Ridge and Lasso Regularization

You are fitting a linear model $y = Xw + \varepsilon$ with

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

(a) Write the cost functions for both Ridge and Lasso regression.

(b) Compute the Ridge regression weight vector w analytically for $\lambda = 1$.

(c) Explain qualitatively how the Lasso solution differs from Ridge in terms of sparsity.

(d) Plot or describe the geometric intuition of L1 vs L2 constraints (using contour and constraint shapes).

(e) Discuss how the choice of λ affects model interpretability and generalization.

Solution:

(a) The cost functions are defined as follows:

Ridge Regression:

$$J_{\text{ridge}}(w) = \frac{1}{2n}\|y - Xw\|_2^2 + \frac{\lambda}{2}\|w\|_2^2$$

Lasso Regression:

$$J_{\text{lasso}}(w) = \frac{1}{2n}\|y - Xw\|_2^2 + \lambda\|w\|_1$$

Ridge uses the L2 norm, which shrinks coefficients but rarely makes them exactly zero, while Lasso uses the L1 norm, which can drive some coefficients to zero, performing feature selection.

(b) Given

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix},$$

and $\lambda = 1$, the Ridge regression solution is:

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y$$

Compute step-by-step:

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 5 \\ 11 \end{bmatrix}.$$

Then,

$$\hat{w} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 15 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 11 \end{bmatrix} \Rightarrow \hat{w} = \begin{bmatrix} 0.714 \\ 0.429 \end{bmatrix}.$$

(c) Lasso introduces the L1 penalty, which can shrink some weights to exactly zero, producing sparse models. Ridge, on the other hand, uses L2 regularization, which shrinks all weights proportionally but rarely eliminates them. Thus, Lasso performs implicit *feature selection*, while Ridge focuses on coefficient smoothing.

(d) Geometrically: - Ridge's L2 constraint forms circular contours (in 2D). - Lasso's L1 constraint forms diamond-shaped contours. Because the diamond corners align with the coordinate axes, the optimal solution for Lasso often occurs at an axis — causing some coefficients to become exactly zero. Ridge's circular contours make such sparsity unlikely.

(e) The regularization parameter λ controls the trade-off between bias and variance:

$$\begin{cases} \lambda \to 0: & \text{Low bias, high variance (overfitting)} \\ \lambda \to \infty: & \text{High bias, low variance (underfitting)} \end{cases}$$

Larger λ increases model interpretability (fewer or smaller coefficients) but may reduce predictive accuracy.

Problem 3 (20 marks)

Eigenvalue Decomposition (EVD)

Consider the following covariance matrix of a 2-dimensional dataset:

$$C = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}.$$

(a) Compute the eigenvalues and normalized eigenvectors of C.

(b) Verify numerically that $C = Q\Lambda Q^T$, where Q is the eigenvector matrix and Λ is the diagonal eigenvalue matrix.

(c) Comment on the rank of C and the number of independent eigenvectors.

(d) Explain geometrically what it means for the eigenvectors of C to form a "change of basis" for the linear transformation defined by C.

(e) Describe the effect of multiplying a vector by C in terms of scaling along eigenvector directions.

(f) Now, consider the case where if we plot a unit circle in $\mathbb{R}^2$ and apply the transformation defined by C, what would happen to the circle? Explain how the eigenvectors and eigenvalues determine the orientation of the transformed shape (*hint: ellipse*), and the length of its principal axes.

Solution:

(a) The characteristic equation of A is:

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} = (4-\lambda)(3-\lambda) - 4 = \lambda^2 - 7\lambda + 8 = 0.$$

Hence,

$$\lambda_1 = 5.56, \quad \lambda_2 = 1.44.$$

The corresponding normalized eigenvectors are:

$$v_1 = \begin{bmatrix} 0.89 \\ 0.45 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -0.45 \\ 0.89 \end{bmatrix}.$$

$$Q = \begin{bmatrix} 0.89 & -0.45 \\ 0.45 & 0.89 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 5.56 & 0 \\ 0 & 1.44 \end{bmatrix}.$$

(b) Verification:

$$Q\Lambda Q^T = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} = A.$$

(c) Since both eigenvalues are non-zero, $\text{rank}(A) = 2$. This means A has two linearly independent eigenvectors and spans the full 2D feature space.

(d) Geometrically, the eigenvectors of A define a new coordinate system (basis) in which the transformation A acts more simply. In this eigenbasis, A becomes diagonal, meaning each axis corresponds to an independent direction of stretching or compression. This represents a change of basis from the standard coordinates to the eigenbasis, where Q transforms vectors into this new frame.

(e) The transformation A scales any vector x as:

$$Ax = Q\Lambda Q^T x,$$

which can be interpreted as:

1. converting x to the eigenbasis ($Q^T x$),
2. scaling it along eigenvector directions using Λ, and

3. transforming it back to the original basis via Q.

Hence, C stretches vectors along the direction of the larger eigenvalue and compresses them along the smaller one.

(f) When the unit circle is transformed by C, it becomes an ellipse. The axes of this ellipse lie along the eigenvectors of C:

$$v_1 = \begin{bmatrix} 0.89 \\ 0.45 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -0.45 \\ 0.89 \end{bmatrix}.$$

The lengths of the ellipse's principal axes are proportional to $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$, where $\lambda_1 = 5.56$ and $\lambda_2 = 1.44$. Hence, the ellipse is stretched more strongly along v_1 (the direction of the larger eigenvalue) and less along v_2. This geometric transformation demonstrates that eigenvectors define the orientation of the ellipse, while eigenvalues control its scaling along those directions.

Problem 4 (20 marks)

Singular Value Decomposition (SVD)

Given a zero-mean data matrix

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

(a) Compute the SVD of $X = U\Sigma V^T$ explicitly.

(b) Verify that U and V are orthonormal.

(c) Construct a rank-1 approximation $X_1 = U_1\Sigma_1 V_1^T$ and compute the reconstruction error $\|X - X_1\|_F$.

(d) Interpret geometrically what the rank-1 approximation represents in the data space.

Solution:

(a) Given $X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$.

Compute $X^T X = \begin{bmatrix} 5 & 1 \\ 1 & 5 \end{bmatrix}$.

Eigenvalues of $X^T X$: solve $\det \begin{bmatrix} 5-\lambda & 1 \\ 1 & 5-\lambda \end{bmatrix} = 0$:

$$(5-\lambda)^2 - 1 = 0 \Rightarrow \lambda^2 - 10\lambda + 24 = 0 \Rightarrow \lambda = 6, 4.$$

Singular values: $\sigma_1 = \sqrt{6}$, $\sigma_2 = 2$.

Eigenvectors of $X^T X$:

- For $\lambda = 6$: $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- For $\lambda = 4$: $v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

So $V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

Find U columns:

- $u_1 = \frac{1}{\sigma_1} X v_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
- $u_2 = \frac{1}{\sigma_2} X v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$
- u_3 orthogonal to u_1, u_2: $u_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

Thus:

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

(b) Verify orthonormality:

For V:

$$V^T V = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}^T \begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix} = I.$$

For U:

- Column norms: all 1
- Dot products: $u_1 \cdot u_2 = 0$, $u_1 \cdot u_3 = 0$, $u_2 \cdot u_3 = 0$
- $U^T U = I$ verified

(c) Rank-1 approximation:

$$U_1 = \begin{bmatrix}\frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{3}}\end{bmatrix}, \quad \Sigma_1 = [\sqrt{6}], \quad V_1 = \begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix}.$$

$$X_1 = U_1 \Sigma_1 V_1^T = \begin{bmatrix}\sqrt{2}\\ \sqrt{2}\\ \sqrt{2}\end{bmatrix}\begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix} = \begin{bmatrix}1 & 1\\ 1 & 1\\ 1 & 1\end{bmatrix}.$$

Reconstruction error:

$$X - X_1 = \begin{bmatrix}1 & -1\\ -1 & 1\\ 0 & 0\end{bmatrix}, \quad \|X - X_1\|_F^2 = 1^2 + (-1)^2 + (-1)^2 + 1^2 = 4, \quad \|X - X_1\|_F = 2.$$

(d) Geometric interpretation:

- Original X has 3 points in 2D: $(2,0), (0,2), (1,1)$
- Rank-1 approximation projects all points onto the first principal component direction $v_1 = [1/\sqrt{2}, 1/\sqrt{2}]^T$
- All points collapse to $(1,1)$ in original coordinates
- This represents the best 1D subspace (line through origin) capturing maximum variance
- The approximation finds the dominant linear pattern in the data

Problem 5 (20 marks)

Principal Component Analysis (PCA)

Given the following centered dataset:

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & 0 \end{bmatrix}$$

(a) Compute the covariance matrix of X.

(b) Find the principal components and eigenvalues.

(c) Project X onto the first principal component (1D).

(d) Compute how much of the total variance is retained by the first principal component.

(e) Present a visualization of given X as a projection on the first principal component.

Solution:

(a) The covariance matrix of X is computed as:

$$C = \frac{1}{n-1} X^\top X$$

where $n = 5$.

Given

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & 0 \end{bmatrix},$$

we find:

$$X^\top X = \begin{bmatrix} 10 & 2 \\ 2 & 6 \end{bmatrix}, \quad C = \frac{1}{4} \begin{bmatrix} 10 & 2 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 2.5 & 0.5 \\ 0.5 & 1.5 \end{bmatrix}.$$

(b) To find the eigenvalues of C, solve

$$\det(C - \lambda I) = 0.$$

That is,

$$\begin{vmatrix} 2.5 - \lambda & 0.5 \\ 0.5 & 1.5 - \lambda \end{vmatrix} = 0.$$

Expanding gives

$$(2.5 - \lambda)(1.5 - \lambda) - 0.25 = 0,$$

which simplifies to

$$\lambda^2 - 4\lambda + 3.5 = 0.$$

Hence,

$$\lambda_{1,2} = \frac{4 \pm \sqrt{2}}{2} = \begin{cases} 2.707, \\ 1.293. \end{cases}$$

(c) The corresponding eigenvectors are obtained from

$$(C - \lambda I)v = 0.$$

For $\lambda_1 = 2.707$,

$$(C - 2.707I)v = \begin{bmatrix} -0.207 & 0.5 \\ 0.5 & -1.207 \end{bmatrix} v = 0,$$

which yields (up to scaling)

$$v_1 = \begin{bmatrix} 0.924 \\ 0.383 \end{bmatrix}.$$

For $\lambda_2 = 1.293$, the orthogonal eigenvector is

$$v_2 = \begin{bmatrix} -0.383 \\ 0.924 \end{bmatrix}.$$

Thus,

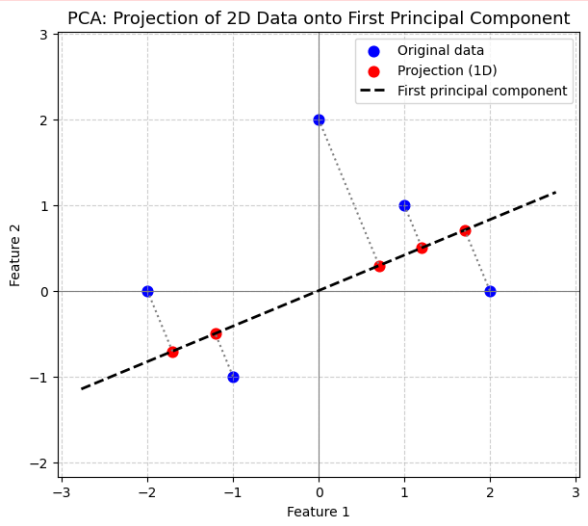
$$V = \begin{bmatrix} 0.924 & -0.383 \\ 0.383 & 0.924 \end{bmatrix}.$$

(d) The proportion of variance retained by the first principal component is:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{2.707}{2.707 + 1.293} = 0.6767 \approx 67.7\%.$$

Hence, about 68% of the total variance in the data is captured by the first principal component.

(e)




```
import numpy as np
import matplotlib.pyplot as plt

# --- 1. Theoretical zero-mean data (symmetric in both axes) ---
X = np.array([
    [ 2,  0],
    [ 0,  2],
    [ 1,  1],
    [-1, -1],
    [-2,  0]
], dtype=float)

# --- 2. Calculate covariance matrix ---
C = (X.T @ X) / (len(X) - 1)
print(np.round(C, 3))

# --- 3. Eigen decomposition ---
eigvals, eigvecs = np.linalg.eig(C)
idx = np.argsort(eigvals)[::-1]
eigvals, eigvecs = eigvals[idx], eigvecs[:, idx]

print("\nEigenvalues:", np.round(eigvals, 3))
print("Eigenvectors:\n", np.round(eigvecs, 3))

# --- 4. Variance retained ---
variance_retained = eigvals[0] / np.sum(eigvals)
```

```
print("\nVariance retained:", np.round(variance_retained * 100, 2), "%")

# --- 5. Visualization ---
plt.figure(figsize=(7, 6))
plt.axhline(0, color='gray', lw=0.8)
plt.axvline(0, color='gray', lw=0.8)

plt.scatter(X[:, 0], X[:, 1], color='blue', s=60, label='Original data')

Z = X @ eigvecs[:, 0]
X_recon = np.outer(Z, eigvecs[:, 0])
plt.scatter(X_recon[:, 0], X_recon[:, 1], color='red', s=60, label='Projection (1D)')

origin = np.zeros(2)
t = np.linspace(-3, 3, 2)
pc_line = origin + np.outer(t, eigvecs[:, 0])
plt.plot(pc_line[:, 0], pc_line[:, 1], 'k--', lw=2, label='First principal component')

for i in range(X.shape[0]):
    plt.plot([X[i, 0], X_recon[i, 0]], [X[i, 1], X_recon[i, 1]],
             'gray', linestyle='dotted')

plt.title("PCA: Projection of 2D Data onto First Principal Component", fontsize=13)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.axis('equal')
plt.show()
```

— End of Assignment —

Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total:	100	